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A game of futures: the strategy of scenarios in a Danish medical company

Andreas Lyse Brandt & Kasper Tang Vangkilde

Abstract

While the future has gained increasing attention in organizational research, the minutiae and intricacies of how organizational actors produce and enact the future remain little explored. In this article, we present an ethnographic study of a future-making practice called 'the strategy of scenarios' in a Danish medical company. We propose the concept of technologies of prefiguration to capture how such practices prefigure the future so as to make it actionable. Our study shows that this unfolds as a 'game of futures' in which prefiguration depends not only on an outward orientation towards societal developments, but equally on an inward orientation towards the political economy of the organization. Although rational reasoning forms part of this game, futures are established and negotiated through *a*rational techniques of enchantment focused on 'selling futures'. This challenges rationalist ideas of future-making, demonstrating the heterogenous, more-than-rational processes and distinct organizational logics by which futures are shaped.

Keywords: future; scenarios; prefiguration; ethnography; medical industry

Introduction

The future figures as a critical concern in most, if not all, contemporary organizations. In a recent publication, Wenzel et al. (2020) argue that prevalent discourses on issues such as climate change, digital transformation, and post-truth politics illustrate the acute preoccupation with the future in organizational life today. This argument is supported by more specific studies that demonstrate how particular constructions of futures (Lê 2013) and the performance of political risk regimes (Nyberg and Wright 2016) shape organizations' situated responses to, for instance, climatic changes. Other studies have

more broadly emphasized that even if much corporate energy is directed towards the future, long-term planning perspectives tend to give way to more short-term future orientations among senior officers (Johansen and De Cock 2018). The future, in other words, has thus been 'rediscovered' (Wenzel et al. 2020) as an inherently indeterminate, open-ended, and uncertain horizon.

Crucially, however, the indeterminacy of the future does *not* mean that efforts to know and foretell it are non-existent. Quite the reverse, in fact. The indeterminacy of the future seems to make necessary, as Luhmann (1976) has argued, 'more anticipatory behaviour – literally, more acting *before* the event' (134, emphasis added). As recently discussed by Flyverbom and Garsten (2021), this is not least evident in contemporary organizations where anticipatory practices of 'foreseeing, foreshadowing, or forecasting future events' (2) have received increasing attention among organizational actors and scholars in recent years. The list of such practices is long and diverse: calculations (Beckert and Bronk 2018b), big data and algorithmic decision-making (Flyverbom and Madsen 2015), business modelling programmes (Thomson and Byrne 2022), fabulation and storytelling (Hjorth 2013), trend and horizon scanning (Juech and Michelson 2012), design and expansive thinking (Skibsted and Bason 2022), workshops and experimental exercises (Bucher and Langley 2016), processes of inspiration (Vangkilde 2015), living labs (Brandt 2023), and more.

For all of these, a key objective is to produce anticipatory knowledge that will bring the future – and what it might entail of 'phenomena, problems and opportunities' (Flyverbom and Garsten 2021, 4) – into the present so as to make it both visible and actionable. Or, as Anderson (2010) points out, 'anticipatory practices […] enable the performative operation of establishing the *presence* of "what has not happened and may never happen"' (783, emphasis in original). Yet, it is a crucial point that 'knowledge of

the future' remains inescapably uncertain and, as emphasized by Nowotny (2016, 7), thus a contradiction in terms. For organizational research, this raises several significant questions, not least concerning the minutiae and intricacies of *how* organizational actors produce and enact the future. How do they seek to know and foretell the yet-to-come in order to act *before* events? Which kinds of expertise and authority become meaningful and gain significance? What are the organizational forms of sociality, subjectivity, and experience that emerge from pursuing anticipatory knowledge?

Such questions have not gone unnoticed in organization studies, although they have hitherto been 'under-appreciated' (Wenzel et al. 2020, 1441) and 'only selectively scrutinized' (Beckert 2021, 1). Particularly in recent years, discussions and studies of futures and their organizing roles and effects have gained increasing currency (Beckert 2016, 2021; Beckert and Bronk 2018b; Flyverbom and Garsten 2021; Johansen and De Cock 2018; Wenzel et al. 2020). Reaching beyond a conception of the future as 'simply the unfolding of events' (Flyverbom and Garsten 2021, 2) or 'an objective "thing" out there' (Wenzel et al. 2020, 1444), a key point in this growing body of research is that futures are constructed, performed, and enacted in the present through particular 'future-making practices' (Wenzel et al. 2020) and 'instruments of imagination' (Beckert 2021) – in other words, through distinct ways and devices by which organizational actors, here and now, seek to know and foretell what is yet to come. Importantly, this performative and practice-oriented approach to futures implies that futures are essentially contested and perspectival, always coming from somewhere, being shaped by specific logics, and imbued with priorities and interests (Flyverbom and Garsten 2021, 5). In a nutshell, far from being 'matters of fact', futures are 'matters of concern' (cf. Latour 2005).

While much conventional organization and management theory have portrayed the process of acting on the future as a rational endeavour, informed by predictive and

rigorous analysis (Gümüsay and Reinecke 2022; Jarzabkowski and Kaplan 2015), such rationalist ideas of future-making are strongly challenged by the perspectives above. In the work of Beckert (2016, 2021), for instance, rationalist (objective, unbiased, factual) and non-rationalist (emotional, performative, fictional) thinking and practice go together in that reason and imagination 'should not be seen as antithetical' (Beckert and Bronk 2018a, 4). Ethnographic approaches employed in recent organizational studies have also shed light on the significance of visual and aesthetic techniques of presentation (Comi and Whyte 2018), spiritual and bodily performances (Mauksch 2017), and moment-to-moment practical knowledge (Thompson and Byrne 2022) in engaging stakeholders and shaping futures. Such studies demonstrate that rationality alone can, by no means, fully capture the intricacies of organizational efforts to manage the future and that rationalist and more-than-rationalist reasonings and processes are intricately entangled. This also hints at how anticipating and acting upon futures are essentially embedded in distinct organizational logics, power structures, and social dynamics. Indeed, as Flyverbom and Garsten (2021) emphasize, we 'tend to disregard the intricate organizational practices and rationalities that go into the production of anticipatory knowledge' (7). To advance this research focus, and to move further beyond a dichotomy of rationalist versus non-rationalist approaches, we contend that more ethnographic research is needed to explore the organizational logics and everyday practices by which futures are brought to life and enacted in organizations today.

Our case in point is the widespread use of scenario planning, which has become a highly influential 'futures methodology' in contemporary politics and organization (Cooper 2010, 170; Andersson 2018; Schoemaker 1995). The scenarios method, Cooper (2010) notes, 'is perhaps the most ubiquitous and most consequential of epistemologies in contemporary politics. Yet it has received surprisingly little academic attention, other

than the most technocratic in nature' (171). Although the literature on scenario planning has grown extensively in recent decades (Varum and Melo 2010), it often has such a normative and instrumental slant that scenario planning research, to quote Balarezo and Nielsen (2017, 3), 'can be described as "Populist Science" where practical relevance is high but theoretical and methodological rigor is low'. Recently, Beckert (2021) has expanded the conceptual framework of scenario planning, emphasizing how scenarios create 'fictional expectations' in organizations; that is, 'convictions on how the future will unfold' (2). He stresses that scenarios not only offer possibilities for exploring various future states, but also influence decision-making and contribute to creating the conditions necessary for their realization (6). Following Beckert, then, to understand scenario planning as a particular approach to the future, it is essential to pursue a research agenda that delves into the specifics of the constitution, contestation, and acceptance of futures in organizations. This will include elucidating the organizational structures, power dynamics, and individual actions which affect how and why future scenarios win or lose credibility (14-15).

In this article, our purpose is to contribute to such a research agenda. Drawing on ethnographic fieldwork in a Danish medical company, here called MedTech [1], we explore how managers and employees approach the future of novel medical treatment by means of what is called 'the strategy of scenarios'. We put forward the concept of technologies of prefiguration as a way to highlight and conceptualize the heterogeneous processes – rational as well as more-than-rational – by which futures are prefigured so as to be openly explored, critically scrutinized, and, not least, acted upon. Our analysis shows that, in MedTech, this unfolds in practice as a 'game of futures' in which the ability to foretell and impact the future of medical treatment not only depends on an outward orientation towards broader medical and societal developments and tendencies,

but equally, or even more so, on an inward orientation towards the particular dynamics of the game; in other words, towards the political economy of the organization. While rational processes and reasoning such as tests and calculations form part of the game, we contend that scenarios are established and negotiated through *a*rational techniques of enchantment focused on 'selling futures' inside the organization. This insight challenges rationalist ideas of future-making, demonstrating the heterogenous, more-than-rational processes and distinct organizational logics and practices in which futures are embedded and by which they are shaped.

But first, we begin by elaborating on our conceptual framework, describing how we conceive of scenario planning as a technology of prefiguration.

Conceptual framework: scenarios as technologies of prefiguration

As a distinct way of approaching the future, scenario planning is closely associated with the intellectual legacy of Herman Kahn, an American physicist recognized as one of the founders of Futures Studies (Aligica 2004, 67). Kahn's ideas were first developed as a military strategist at RAND Corporation, an American non-profit think tank created in 1948 with the purpose of providing research and analysis to the US Armed Forces. In response to the increasing tensions between the United States and the Soviet Union after World War II, new techniques of 'preparedness' were needed to anticipate the threat of a nuclear attack (Lakoff 2007, 255-256). In the 1950s, the RAND strategists criticized military planners for their approach to the tenets of deterrence, the very principles of the Cold War. As Lakoff notes (2007), Kahn was prominent in this critique: 'If planners were serious about the strategy of deterrence, he [Kahn] argued, they had better be prepared to actually wage nuclear war' (259). The problem, as Kahn saw it, was that military planners had no specific understanding or vision of how a nuclear war would actually unfold.

In Kahn's view, this lack of preparedness and planning for unprecedented events was a problem not only in the US Department of Defence, but in strategic thinking in organizations more generally. Kahn therefore created the Hudson Institute in 1961, a conservative think tank with the aim to sophisticate interdisciplinary examinations of 'topical problems that have a significant future-oriented dimension' (Aligica 2004, 72). Kahn believed that the usual output of think tanks – that is, reports with unambiguous conclusions – were insufficient as knowledge production about future-oriented issues. The problem, as he saw it, was that the assumptions, criteria, and calculation techniques underlying conclusions in reports are very uncertain. Instead, such explorations should have a pedagogical and heuristic finality (Aligica 2004, 72-73). Kahn's solution, then, became to construct scenarios. Together with his colleague, Anthony J. Wiener (1967), he defined scenarios as follows:

> Scenarios are hypothetical sequences of events constructed for the purpose of focusing attention on causal processes and decision-points. They answer two kinds of questions: (1) Precisely how might some hypothetical situation come about, step by step? and (2) What alternatives exist, for each actor, at each step, for preventing, diverting, or facilitating the process? (6).

To Kahn, in other words, a scenario is not a predictive device. Rather, it is a systematic and methodological effort to generate, through analysis and imagination, a number of alternative futures and the possible paths that may lead to those futures. The gains of such an approach are that it allows for exploring, testing, and evaluating possible and unknown futures; that is, 'one may see better what is to be avoided or facilitated, and one may also gain a useful perspective on the kinds of decisions that may be necessary' (6). Scenarios, thus, constitute an 'ontology of possible worlds' (Aligica 2004, 76).

With Kahn and his colleagues being the visionaries who introduced the focus on preparedness through scenarios in public policy and organization, one corporation, in

particular, popularized scenarios more broadly. In the 1970s, the Royal Dutch/Shell Group distinguished itself by coming out strong from an increasing turbulence in world economic affairs, not least caused by the international oil crisis. The well-known story is that the company had applied a particular planning technique which had prepared it to such an extent that the oil crisis did not really come as a surprise. A key person was the economist Pierre Wack who, like Kahn, believed that the future cannot be predicted and that forecasting techniques are accordingly a waste of time and money. In two articles in the *Harvard Business Review*, Wack (1985a, 1985b) reflects on Shell's development of a planning technique to anticipate an uncertain future: 'The future is no longer stable; it has become a moving target', for which reason it is essential 'to accept uncertainty, try to understand it, and make it part of our reasoning' (1985a, 73). To this end, the crafting of scenarios is a productive method in that it entails a 'willingness to face uncertainty and to understand the forces driving it' (74).

Over the years, scenarios have since become highly influential in politics and organization as a response to 'turbulent worlds' (Cooper 2010). Despite recent efforts to bring theoretical rigour to scenario planning (Andersson 2018; Beckert 2021), previous research has hardly moved beyond instrumental and technocratic perspectives (Balarezo and Nielsen 2017; Cooper 2010). This has left us largely without nuanced empirical explorations and understandings of how scenarios are crafted, performed, prototyped, negotiated, contested, and decided by specific social actors in specific organizational contexts; that is, if you will, the 'everyday organizational life' of scenario planning in practice.

Such a focus resonates with a more general interest among anthropologists who, in recent years, have paid increasing attention to the future, exploring how people relate to, and act on, possible and unknown future worlds (see, e.g., Appadurai 2013; Nielsen

2011). An intriguing aspect of these studies is that people's actions tend to be informed by a vision of what has not and, indeed, may not happen. 'The present', as Nielsen (2011) contends, 'thus becomes a function of an imagined future moment which is extended backwards in time to ground the current act' (398); that is, otherwise detached temporal moments become interwoven as the future is rendered present *in* the present.

While this emerging 'ethnography of the future' has led some anthropologists to explore new ways of engaging actively in future-making (Salazar et al. 2017; Smith et al. 2016), others have taken a more distanced and critical approach which uncovers the cultural imaginaries and socio-political processes involved in attempts to design future worlds (Suchman 2011; Vangkilde 2017). Regarding the concern with climate change, for instance, the latter approach has shown how climate change modelling, as a distinct mode of anticipating nature, is socially embedded and no neutral act; that is, arguments, interests, and politics are at work through and through (Hastrup and Skrydstrup 2013). As forms of scenarios, climate models exert power and authority in the present as they frame and shape how people perceive and experience the world and how they plan and act in it. Models and scenarios thus infiltrate human life and existence, and it remains a key task for researchers to ethnographically disclose the practices and logics underlying their creation.

It is to this end that we propose the concept of technologies of prefiguration as a way to highlight and conceptualize the heterogenous processes through which possible futures are created, scrutinized, and acted on. In our conceptualization, technologies of prefiguration denote established, systematic, and, in many instances, institutionalized knowledge processes that serve to *pre*figure the future in order to make it actionable. In organization studies, the concept of prefiguration has mostly been used to examine how organizational actors, shaped by individual and collective actions rather than ideologies,

get involved in, for instance, political movements (Farias 2017), environmental activism (Skoglund and Böhm 2020), or redefining work communities through self-management (Kokkinidis 2015) [2]. It is worth noting, however, that prefiguration is most commonly associated with the more specific notion of 'prefigurative politics', which is connected with various social movements. Here, it refers to political activities where the temporal distinction between a future goal and the present means to achieve that goal is dissolved or conflated (Maeckelbergh 2011, 4); or, in other words, the political ends are expressed or acted out through their means (Yates 2015, 1). As Yates (2015) thus explains, 'to prefigure is to anticipate or enact some feature of an "alternative world" in the present, as though it has already been achieved' (4). Building on these studies, technologies of prefiguration emphasize the practical enactment of futures in the present and the ways in which they, through their performativity, serve to not only explore possible scenarios, but also to actively work towards their realization [3].

By using the concept of technology, we follow a number of anthropologists who, by drawing on Foucault and his deliberations on governmentality and biopolitics (2007; 1991), explore the mechanisms by which organizational actors prepare for and govern the future (Flyverbom and Garsten 2021; Collier and Lakoff 2015; Samimian-Darash and Rabinow 2015) [4]. With an emphasis on uncertainty-based governmental technologies (Samimian-Darash and Rabinow 2015), these anthropologists draw upon a Foucauldian notion of technology that refers 'not to tools, machines, or the application of science to industrial production, but rather to methods and procedures for governing human beings' (Behrent 2013, 55). In a recent article on anticipation and organization, Flyverbom and Garsten (2021) likewise draw on Foucauldian perspectives as they argue that anticipatory governance within organizations is intricately tied to the production of knowledge. Here, knowledge is far more than mere information in that it encompasses

specific resources and practices as well as power dynamics. Knowledge, they contend, 'entails organizing resources in a way that has an *effect*' (5, emphasis in original); which is to say that it intervenes in social relations by way of governing others, shaping their attention, conduct, and subjectivities (4-5). Accordingly, futures are not 'to be observed and reported on at a distance' (5), but constructed and perceived through knowledge processes that are inherently temporal, perspectival, and performative as central features of 'efforts to shape conduct' (5).

Technologies of prefiguration, thus, constitute organizational forms of knowing that, through the practical and performative enactment of futures (such as scenarios) in the present, underpin and foster strategic efforts to govern the future. As *technologies*, they fundamentally impinge on and shape the realities that become the everyday worlds we inhabit. This indicates the power relations and politics involved in a Foucauldian sense; that is, the prefigurations emphasize certain aspects and neglect others and, thus, encroach on human worlds and conduct in particular ways. Although Foucault used the terms 'technology' and 'technique' interchangeably (Behrent 2013, 59), in this article, we distinguish between them as technologies serve as an overall concept to highlight the effects of prefigurations on social relations and human conduct, while techniques refer to the specific practices and strategies by which such prefigurations become established and gain or lose credibility in organizations. As such, we seek to add to the emerging focus on futures in organization research by proposing technologies of prefiguration as an analytical tool to expose and tease out how future scenarios are deeply entangled in social and political relations. To substantiate our argument, we draw on ethnographic fieldwork conducted in the medical company MedTech.

Research method: ethnographic fieldwork in MedTech

MedTech is a Danish medical company located in the eastern part of Denmark. While it has lately been through a turbulent time with a number of serious cost reductions, not least a process of laying off hundreds of employees, the company has experienced solid economic growth for years. With a focus on medical treatments for patients with serious and chronic diseases, the market for the company is in constant need of new and better health solutions. A quest for innovation thus constitutes a central working condition in an industry that is stretched out, so to speak, between an everyday search for improved healthcare and a product development process spanning years, if not decades, due to its complexity and rigor. For the more than thousand employees dispersed in facilities in both Denmark and abroad, the future has, however, looked much brighter in the past than in 2016 when one of us, Andreas, carried out an ethnographic fieldwork in MedTech. At that time, the future posed first and foremost as a looming uncertainty because of a series of critical developments and new restrictions on the most important of MedTech's regional markets.

In MedTech, innovation and design unfold within a strategic framework called 'the strategy of scenarios'. All innovation projects are conceived as scenarios that move through a series of stages and milestones from A to I covering all steps from the initial idea to the product launch. When the management announces a new strategic focus on, for instance, digitalization, the employees are given free rein to propose new ideas for innovation projects. To qualify as a scenario, however, an idea must first be accepted by the management after which it forms part of typically three-four 'scenario candidates' related to the specific innovation area. During the stages from A to C, potentially lasting up to three years, the scenarios are discussed and evaluated at predefined milestones that concern such issues as business opportunity, design concept, project feasibility, and

patient perspectives. At stage C, a final decision is taken regarding which scenario will be the 'leading candidate', which means that it will proceed to the next stages of more specific development (D to G) and marketization (H to I). When a leading candidate is selected, the second-best scenario is typically put on hold while the other scenarios are closed down.

This strategic framework is, for instance, operative in Tech R&D that represents one department out of seven in the Research and Development (R&D) Division. While most of the other departments are concerned with inventing new or improved medical drugs, the purpose of Tech R&D is to develop the technological devices to be used with the drugs. The employees in Tech R&D thus consist mainly of engineers and designers, although other professionals such as business strategists, usability researchers, human resources personnel, anthropologists, and managers are also present in the two-story building that houses Tech R&D at a large facility on the outskirts of a provincial city.

For six months in 2016, Andreas joined the department's anthropological team as an intern, his main tasks being to assist with conducting ethnographic studies on how people live with a demanding disease. Since 2014, two anthropologists have been part of Tech R&D with the purpose of contributing to the company's declared ambition to bring in-depth insights about patients' everyday lives into the design of new products. The anthropological work thus consists largely of two parts: one focused on generating knowledge through ethnographic research and the other on disseminating and activating this knowledge in the organization. Due to his specialization in design anthropology, Andreas was offered an internship in Tech R&D with the objective to learn from, and participate in, the work of the anthropologists.

This position allowed Andreas to explore the organizational processes by being 'in the midst of things' (Gatt and Ingold 2013, 148). While this is a general quality of

participant observation, Gatt and Ingold (2013) reverse the wording and put forward the notion of observant participation to emphasize how a fieldworker may not only learn from the interlocutors in the field, but also contribute actively to the activities that unfold. Observant participation, they emphasize, 'takes participant observation one step further' (154). This is an apt description of the approach adopted by Andreas who took part in the preparation of field studies, analysis of data, presentation of results, writing of reports, and mundane activities such as meetings, coffee breaks, and small talk. As ethnographic research always rests on a personal engagement, observant participation takes this seriously by highlighting how a fieldworker seeks to become a participant *in*, not merely an observer *of*, the lives of the fieldwork subjects.

The role of observant participant presented both methodological opportunities and challenges. As has been stressed by several organizational ethnographers, much of organizational life is made up of meetings as constitutive social forms (Schwartzman 1989; Sandler and Thedvall 2017). In MedTech, meetings likewise represent significant social interactions that require employees to leave their desks and gather in dedicated meeting rooms, thus punctuating their activities in the otherwise open office landscape (Krause-Jensen 2010, 29). Andreas was not only granted access to numerous meetings, but also gradually learned to leverage his participation in them as a way to purposely generate empirical insights. Often, the meetings themselves transformed into informal and unstructured interview-like interactions, sparked by Andreas' curiosity and 'naive questioning'. In one meeting, for instance, a manager thus diverted from the meeting topic to articulate his strong views on the 'political interests' surrounding innovation in MedTech. At the end of the meeting, he added that what he had just said had to remain confidential, thus creating an aura of mystique that hinted at the importance of scenarios and their impact on the social dynamics in the organization. Meetings – and Andreas'

observant participation in them – constituted, in this way, arenas for organizational life in general and the exploration of scenarios in particular.

While taking active part in various organizational tasks and meetings, Andreas encountered, however, a dilemma of being deeply immersed in the field and upholding analytical distance (Ybema and Kamsteeg 2009). More specifically, he faced a risk of 'overthinking' – that is, of reading his own ideas into concepts and accounts that appear deceptively familiar (Vangkilde and Sausdal 2016) – as he did ethnographic fieldwork among people who, if not even being anthropologists themselves, represented what Holmes and Marcus (2008) term para-ethnographers; a concept developed to capture the reflexive practices and ethnographic curiosity existing among people in many field sites today (see also Vangkilde and Rod 2016). In line with Holmes and Marcus' argument, this inspired Andreas to engage actively with the employees and managers as 'dialogic, epistemic subject[s]' (Holmes and Marcus 2008, 85), who not only articulated their own reflections and analyses, but moreover responded to, and often challenged, Andreas' insights and analyses. In these encounters, overthinking could be resolved and analytical distance upheld through open dialogue and critical debate as an significant part of the observant participation (Vangkilde and Sausdal 2016).

Importantly, this engagement involved contacts and relations with many other employees than solely the anthropologists. Andreas could thus complement his use of observant participation with qualitative interviews with different agents involved in the strategy of scenarios. Concretely, he recorded eleven semi-structured interviews with senior managers, scenario project leaders and members, a business strategist, and the anthropologists to cover different perspectives in the organization on how the strategy of scenarios is enacted. A large number of more informal conversations, characterized by spontaneously occurring during or after participation in activities and meetings, also

form part of the empirical material. Compared to the observant participation, interviews and conversations allowed for distanced questioning and reflection on the processes of prefiguring and acting upon an indeterminate future. Combining engaged participation, dialogical encounters, and distanced questioning thus served to explore how the strategy of scenarios was performed and negotiated by managers and employees in MedTech.

Ethnographic findings: the strategy of scenarios in practice

As already noted, the strategy of scenarios constitutes a fixed model of innovation that profoundly impacts the everyday work. For many employees, the scenario candidates determine not merely what they are working on, but also how they are organized into project teams. The strategy moreover entails a deep uncertainty regarding the endpoint or destiny of each scenario; that is, everyone knows that a scenario may turn into the next big hit for the company or, more likely, that it will eventually be rejected. 'There are many employees who have worked in this company for more than ten years without having launched a single product yet' was a popular saying in Tech R&D, as if to stress the inherent uncertainty and logic of the scenario strategy.

At the time of Andreas' fieldwork, a deep concern was emerging in MedTech concerning how digitalization would make a profound impact on the future of pharmacy and illness treatment. Referring to various forms of digital health solutions such as large databases, technological devices, and tools for apps, an innovation area called Digi-Pharma was established by the management. In an interview, a manager explained how digitalization was an unknown domain of business for the company:

> All these digital possibilities, tools, and platforms are moving closer and closer to the world of pharmacy. I don't think that we have quite figured out what to do with Digi-Pharma […] But you can say that if you are *not* engaged in it and *don't* understand it, then you are not prepared for the future.

According to the manager, digitalization 'is a train that moves fast with an entirely new competitive situation that we have to consider'. This situation, along with its inherent uncertainty, was also felt by the employees, some of whom experienced it as somewhat paralyzing. In an interview, a scenario project leader explained: 'On a previous project, I was highly active in figuring out what this digital feature could look like. Now, I'm just like… I don't even want to dig into those discussions anymore, because it's all a matter of decision-making on the highest level in the organization'. As these interviews illustrate, Digi-Pharma – or, more broadly, the emerging digitalization of the medical industry – constituted a 'present future', in Luhmann's words (1976, 142), that figured as a kind of projection screen for both business opportunity and economic threat in the company. The scenario candidates within Digi-Pharma may, in this sense, be seen as different possible directions into the future, which, through their prefiguration, make the uncertain future of digitalization available for exploration and scrutiny.

In what follows, we dive into the ethnographic analysis of how this unfolded in practice. We shall show how the endeavours to prefigure the future is no trivial act, but rather amount to a kind of game in which various actors try hard, if not even struggle or compete, to specify the future and stake out the direction into it. While the game thus involves a number of differently positioned players with competing visions, ideas, and dispositions, we seek to demonstrate that techniques of enchantment matter just as much as, or even more than, rational reasoning. Or, in other words, the game is 'won' by those who succeed in making what some actors in MedTech call 'a sale'; that is, strategies of appeal and persuasion are at the core of the game of futures.

Making allies and the importance of timing

Formally, a scenario project begins with a document called an 'innovation brief'. This document must contain information on predefined factors such as the technological and

design proposal, targeted user needs, estimated production costs, and imagined profits. Basically, the document outlines an initial business case and the contours of a possible future, which should enable the management to assess both the opportunities and the risks involved. It is, however, by no means that straightforward. As an engineer with more than 20 years of experience in MedTech stated in an interview:

> You can't describe the essence of a project in tables; it just becomes numbers. You simply can't sell projects on a piece of paper alone […] Then, you'll never create that moment where one can say: 'Yeah okay, there is a leak here, but on the other hand there is so much greatness there'.

This engineer knows what he talks about. During his many years at MedTech, he has been highly successful in turning ideas into scenarios, giving him quite a few product launches on his CV. To him, it is clear that numerical tables and rational reasoning will only take you so far. While the innovation brief is a fixed requirement, something else is at play, too. Something equally, or perhaps even more, decisive.

Within the domain of Digi-Pharma, a project called Rethink quickly went from an initial idea to a scenario candidate. Three employees working on Rethink took on the challenge of digitalization and creatively combined elements from two earlier scenario candidates that had recently been closed down within a different innovation area. This had made the three employees feel, as one of them said, 'relegated to the second-best league and eager to return to the Premier League'. Thus, they now made a crucial move. When asked about how Rethink became established as a scenario candidate so quickly, one of them described in an interview:

> We kind of felt that we were standing on a burning platform, but also that we had a really good idea of how to get away from it. So, we decided to do a mind map focusing on 'okay, who can we get on board, and who should we convince that this is a good idea?'

A few months later, Rethink had become the far most hyped project of the Digi-Pharma candidates. Another engineer outside the project related that this was due to the team members' ability to create their concept with a bold approach and careful timing. 'Some might say that the Rethink team has poached ideas from other projects', the engineer explained to Andreas one day during lunch. 'But you got to give it to them. They have really tailored their story and timed it perfectly'.

The Rethink team member quoted above agreed that a certain sensibility towards timing and allies is a key factor. As he said: 'We have just managed to reach out to the right persons at the right time'. Crucially, this is not an aspect related to Digi-Pharma only. Another member of the Rethink team, a female designer, emphasized in another interview that 'you simply can't achieve anything in this company just by sitting on your chair'. Instead, she elaborated that you have to act or perform in a particular way:

> There are *so* many times where I have just spoken with someone for five minutes, discussed this or that, and thereby received inputs to the project. You know, that's simply what this work is all about. And some people are more skilled at doing that than others.

With this ability to approach the right persons at the right time, the Rethink project team managed to create a fuss and interest in the potential of the Rethink idea. But more than simply timing and making allies are of crucial importance. We turn now to the power of material prototypes.

Materializing time and the power of prototyping

The employees above stress the importance of a mode of acting that implies a distinct kind of engagement and performance. '*Show it, don't tell it!* is simply alfa omega in this organization', an engineer explained in an informal conversation. This, however, is not only a matter of approaching the right persons at the right time in order to get feedback

and engage in persuasive dialogues with them. Furthermore, it is about drawing on the active agency of material things, on their potential power as evocative objects that may induce emotion and provoke thought (Turkle 2007).

In Tech R&D, this is immediately manifest when one takes a tour around the two-story building. Material prototypes of devices are visible everywhere, lying on desks, showcased in displays, or visualized on posters. In the printing room, a container stores hundreds of disposed prototypes which clearly illustrate the influence that prototyping has in processes of innovation. At one point, a project team even decided to put up large posters of their prototype in order to promote their specific scenario. 'It's almost like election posters in a campaign', a designer said with a smile in an interview. Prototypes are thus used strategically, as was also evident during a meeting between an engineer, the two anthropologists, and three representatives from an external company. The purpose of the meeting was to explore the possibility of proposing a new scenario project to the management. At some point during the meeting, the engineer stated:

> The management is constantly presented with different projects. If we don't show them an image – and I literally mean an image – of a project, then we will miss the opportunity because words alone can't describe the vision. PowerPoints just don't sell!

In the making and justification of scenario candidates, materiality and visuality are of great importance. The same engineer made this perfectly clear when he, in a follow-up interview, elaborated on his statement during the meeting: 'If you can get a thing into people's hands so they can interact with it and get that feeling of "I would love to own this", then it's a whole different story. *Then* you have a sale!'

This strategic use of prototyping supports the argument made by Jiménez (2014) that prototyping has 'become an important currency of explanation and description in art-technology contexts, where the emphasis is on the productive and processual aspects

of experimentation' (381). As porous and indefinite (383), prototypes may be seen to democratize innovation as their open-ended and relational characteristics enhance debates about the making of the future. A prototype never quite accomplishes its own closure (385), meaning that it 'appears as a figure of possibility and suspension' (383). Importantly, however, a prototype must possess certain characteristics to do its work. Not everything goes, of course.

This was evident in an ideation workshop with engineers and anthropologists. The workshop formed part of the initial stages of creating a scenario project. One of the anthropologists opened the workshop by stressing that an idea had to be turned into an innovation brief in order to obtain funding for further development. At this point, an engineer stated promptly: 'It's all about politics! The project *must* address something bigger, some hot topic trending in the company at the moment'. This led to a discussion about how several managers were currently preoccupied with adherence and motivation among patients, so this issue would be good to explore further. Both the engineers and the anthropologists agreed that what was needed was a simple and tangible prototype that would address and manifest a value proposition acknowledged by the managers. 'We have to get into the money stream', the engineer stressed during the workshop. 'So, the first prototype should be used as a showcase that relates to a broader concept which may eventually be sold to the management'.

The prototype, in other words, cannot be of just any kind. It has to resonate with the managers in the sense of responding to, or matching, some kind of emerging present that the managers already sense, although perhaps only partly. In this way, the prototype serves to materialize a distinct temporal moment which is not the present as we know it, but also not an entirely distant future. It occupies a temporal space in-between present and future, both confirming and pushing ahead the managers' ideas and perspectives. 'It

is totally trivial', an engineer said in an interview, 'but you need to have the right thing available at the right time'. While the prototype thus speaks into a current concern, it also allows the employees to articulate, as another engineer put it in an interview, 'hey, look at what we can make if we want to!' The power of prototyping, then, consists in materializing time and affording new understandings that are open-ended and indefinite yet tangible and material. Only then, as we have heard already, 'you have a sale!'

Evaluating scenarios and the vectors of concern

With the employees working on and proposing various scenarios, the question remains as to how the managers tell 'the winners from the losers'. How is it possible to evaluate and decide on matters that are yet to, or may never, happen? To this end, the managers' evaluation and decision-making include a range of assessments and tests that serve to explore each scenario in detail. Evidently, this is not something that the employees take easily. A project member of a Digi-Pharma candidate explained during a conversation:

> There was this desirability test in which all candidates were tested on pre-selected users who would then decide the poorest device, the best device, and so forth. I tell you, when representatives from all candidates planned for this test, everyone went into every last detail!

In addition to the desirability test, production simulations and expert statements are also considered. The anthropologists, for instance, are frequently asked to provide input on various scenarios based on their anthropological knowledge of patients' everyday lives with chronic diseases. Each scenario is also simulated in the company's existing value chain of production, distribution, and retailing infrastructures. The tests, assessments, and simulations are then gathered and presented to the management team by a strategic analyst in charge of facilitating the decision-making process. The analyst described in an interview:

> Each scenario is assessed on the basis of particular parameters and then put into a *pugh matrix*, which is a way to qualify different options against one another by inserting test results relating to specific criteria. But you have to explore each criterion in depth, of course. What are the production costs for each scenario? Are there any possible production complications? Are there technological risks? What do patients prefer? Questions like that.

A broad range of criteria – from desirability tests and anthropological knowledge over production simulations and economic costs to technological possibilities and business opportunities – are thus scrutinized and evaluated. In this process, the scenarios enter a state of 'future presents' (Luhmann 1976, 140), meaning that they are enacted as if they were present realities. This enables strategic analysts and, eventually, managers to make assessments that give insights into 'what kind of reality is emerging if we go with this or that scenario', as a manager said in an interview. Hence, the purpose is to reduce the complexity and uncertainty of the future – of Digi-Pharma in this instance – into more calculated and 'known' risks and dependencies that can be both acted upon and selected between.

An important issue is, however, that tests and other assessments do not speak for themselves. The different scenarios may turn out to be fairly equal in terms of risks and opportunities, thus making it necessary to prioritize between criteria. Is a technological innovation worth pursuing if the business case seems risky? Are low production costs more important than high patient preferences? Etc. The strategic analyst described that managers in Tech R&D typically go for the scenario with the lowest technological risks and the highest patient preferences. In other divisions of MedTech, other criteria such as low production costs or added brand value may be prioritized. Different criteria, in other words, constitute different 'vectors of concern', understood as distinct perspectives that might be taken up and deemed significant in certain situations (Vohnsen 2015).

It follows that evaluations of scenarios are always dependent on negotiation and context – and, by implication, quite competitive. According to one of the managers, this has a particular benefit: 'The competition between the scenarios is very healthy because people need to make an effort'. Clearly, this is a managerial viewpoint, but an engineer was inclined to agree: 'People are extremely dedicated towards their scenario project. Everyone tries to turn their project's distinctive characteristics into the decisive ones. I just love that dedication!', the engineer said. It is part of the story, however, that the competition can also be harsh when a scenario project is closed down. 'It may become really draining', another engineer admitted in an interview, 'if you feel that whatever you do and how well you do it, it just gets turned down'. This finally brings us to a fairly paradoxical matter; namely, how (seemingly) failed scenarios are celebrated in Tech R&D.

Celebrating 'failures' and the logic of defuturing

On a Monday morning, all employees in Tech R&D were gathered in the canteen. One of the purposes of the gathering was that the director of the department would announce which scenario of a group of four candidates had been selected as the leading candidate related to Digi-Pharma. One by one, the director began to announce which scenario had been *de*selected, emphasizing how the exploration and assessment of each scenario had revealed important insights into possible risks and difficulties that had effectively led to the project being closed down. Significantly, he closed each announcement with these words: 'I really think that we should give this project a hand because they have made a great job'. A loud applause filled the canteen, and so it continued for each deselected scenario until only the leading candidate remained.

Seen from the outside, there is clearly something peculiar about this event. At first glance, it seems as an organizational celebration of failures. A deselected candidate

is a scenario deemed either unprofitable, unlikely, unsustainable, unclear, or perhaps even a combination of these and other criteria. In brief, it is a distinct future that turned out to be 'not right' after all. Or so, at least, it seems. For the celebration of seemingly failed futures is a key component of the strategy of scenarios, which is evident by the fact that the deselected futures were celebrated in the canteen with an excitement that equalled, if not even surpassed, the selected one.

Although all employees are dedicated to make an impact on medical treatment, it is thus possible to work at Tech R&D for years without having a scenario selected. Both managers and employees emphasize the counterintuitive point that deselected scenarios provide the same value as selected ones. Although competition is part of the work, it is also stressed that winning or losing the selection of scenarios is not really what matters. As a team leader made clear in an interview:

> The deselection of a scenario may give important insights that make the choice of another scenario more certain. With a deselected scenario, we have explored one possible path. What would have happened if we continued down this road? What could we have met? It is precisely this exploration that makes the selection of a scenario an informed choice.

The strategy of scenarios thus rests on what may be termed a logic of defuturing. The concept of defuturing has been applied, for instance, by Fry (2009) and Hunt (2011) to criticize how contemporary design is unsustainable [5]. Here, however, we draw upon Luhmann (1976) who has written about defuturing as a social dimension of deselecting possibilities made necessary from the conception of the future as an open horizon of the present. As already stressed, Digi-Pharma represents, to use Luhmann's words, a 'future that cannot begin' (141) in the sense of being an infinity of not-yet-realized possibilities that are generating complexity and uncertainty for managers and employees in Tech R&D. It is this complexity that makes necessary a technology of prefiguration such as

the strategy of scenarios. The strategy operates in an inherently defuturizing way; that is, by deselecting and destroying possible futures one by one, complexity is perceived to be reduced and the selected future figures with force and certainty. In this way, what appears as a failed or defuturized scenario is, in fact, not a failure in any absolute sense of the term.

Discussion: the game of futures

The ethnographic findings clearly demonstrate how a profound concern with the future pervades organizational processes. The strategy of scenarios constitutes a technology of prefiguration that brings possible futures into the present in order to govern and act on the complexity and uncertainty of future medical treatment. While it may be articulated in MedTech that winning or losing is not essential, it is at the same time evident how the employees put much effort into generating and promoting their particular scenarios. Far from being a smooth and rational affair, we contend that this unfolds as a distinct game with different players who strategically propel their prefigurations of the future by way of a fine-tuned inward sensibility towards the political economy of the organization and through various techniques of enchantment. We elaborate on this in the following.

An inward sensibility and the political economy of futures

A key characteristic of organizational strategies for anticipating and acting on the future is their general *outward* orientation. The point of such strategies is to look beyond the organization in question so as to generate knowledge, however partial it may be, on how a given market or area of interest may develop. Whether such strategies draw upon big data to uncover broader trends and developments or thick data to explore specific needs and preferences, they are oriented outwards in pursuit of *temporal* perceptions of what

is yet to happen. Indeed, the strategy of scenarios is no different as it renders present a number of possible forthcomings in the field of health and illness.

What our ethnographic findings demonstrate, however, is that such temporal perceptions are deeply embedded in the *spatial* landscape of the organization; that is, to effectively prefigure futures requires a fine-tuned *inward* sensibility towards what Flynn (2009) has conceptualized as 'the political economy of design'. With this notion, Flynn draws attention to 'the ever-shifting landscapes of ideologies, power, and competitions between individuals and groups within the sites of design and development that directly or indirectly define what will be created for consumption by the imagined users' (54). Possible futures, in other words, emerge not merely through data analyses and creative imaginings, but are essentially *performed into being* through socio-political exchanges and strategic positionings that, in turn, also impact how possible futures are perceived and shaped. The production and enactment of futures thus unfold as a locally embedded organizational game, in which the ability to prefigure the future depends on a fine-tuned sensitivity towards the internal dynamics of a political economy of futures.

The conceptualization of social landscapes as games is famously associated with Bourdieu's (1977; 1990) theory of social practice. In this theory, Bourdieu emphasizes how human agents are embedded in socially structured fields in which they, through their practices, come to embody specific social structures as dispositions that give them a practical sense or 'feel for the game' (1990, 66). This sense entails not merely that everything in a given field is perceived to make sense, but also that one has a capacity for anticipating the forthcoming; that is, to quote Bourdieu (2000), 'the good player is the one who […] places himself not where the ball is but where it is about to land' (208). The forthcoming is thus 'already there in the configuration of the game and in the present positions and postures of team-mates and opponents' (208).

In Tech R&D, one part of this capacity concerns the outward orientation towards future medical treatment. However, this part will take you nowhere if it does not relate to, and resonate with, another part – the inward sensibility – which enables an employee to know which strategic actions to take at what time. In the political economy of futures, actors need to be able to play the organizational game and, in Bourdieu's terminology, sense where the ball is about to land in the specific configuration of players, positions, concerns, and visions in the organization. Importantly, this requires competences and techniques which, by no means, can be reduced to mere rational reasonings and actions.

Beyond rationality and the technique of enchantment

The analogies to a game and, by some in Tech R&D, to a sale are not random, but hint at the crucial importance of persuasion or, more precisely, enchantment. As Woolgar (1998) argues in a discussion of innovation, '[s]omehow, somewhere, someone has to be convinced that it is a "new idea"' (442). Similarly, the work on scenarios is deeply socially embedded, being not only a question of creating technological novelty, but of mastering different techniques of enchantment by which others, first and foremost the management, may come to share the belief in the 'presence' and 'rightness' of a distinct possible future.

The concept of enchantment finds its roots in the work of Weber (1930), who discussed the 'disenchantment of the world' as a central characteristic of modernity, reflecting the devaluation of religious, magical, and animistic beliefs about nature and reality (Jenkins 2012). In the field of cultural production and the anthropology of art, Gell (1992) emphasizes that the enchantment of artworks arises not from their physical aesthetics or visual effects, but from the way they are perceived as having come into existence. The power of art objects lies in 'their becoming rather than their being' (46). Building on this idea, Hastrup (2001) argues that the enchantment of a theatre play rests

largely on the performers' ability to master a set of techniques by which they convince the audience about the reality on stage, thus creating 'the illusion of "presence" in this other world' (36). Theatre, in this way, is not faking, but making (41).

Although art and theatre may appear different from medical futures, the concept of enchantment has also been applied in organization studies to highlight how certain managerial acts captivate and motivate specific actions and visions (Hancock 2005) [6]. Mauksch (2017), for instance, conducted ethnographic fieldwork in and around social entrepreneurship events to examine how actors at these events enact compelling visions of social entrepreneurship on stage, thus aiming to inspire, motivate, and create lasting effects in their audiences. Mauksch argues that enchantment emerges as a balancing act between a deeply held belief in a vision, accompanied by investments in aesthetics, creativity, and play, and the need to strategize that vision through reflection, analysis, and control (148).

Techniques of enchantment are likewise at work in Tech R&D, however not just among managers, but equally among employees. In the game of futures, the employees' abilities to timely sense and materially prototype futures, as well as to make alliances around them, function as techniques of enchantment by which the presence of possible futures is produced. This challenges dominant rationalist ideas of scenario planning that stress the objective of enhancing organizational learning about the future by building, testing and deciding between possible alternatives (Balarezo and Nielsen 2017). In Tech R&D, the fixed steps, briefs, decision points, and assessment criteria illustrate how such rationalist reasoning also underlies the strategy of scenarios. This form of rationality, however, is only effective on the surface; behind the surface, a number of techniques of enchantment prevail in the game of futures. These techniques are neither rational nor irrational. Rather, they qualify as what Jöhncke and Steffen (2015) term *a*rational; that

is, they constitute culturally appropriate and practically sensible practices which are not necessarily explicitly legitimate (32-35).

Overall, the techniques show how the prefiguration of futures in Tech R&D rests on a delicate balancing between a temporal perception of the forthcoming and a spatial sensitivity towards the organizational landscape of ideologies, power, competitions, and alliances. Thus, to paraphrase Ardener's (1989) meditations on prophecies, the creation of scenarios links language, time, and space, to which we may add materiality in terms of evocative prototypes. This means that to 'sell' an idea into a scenario, and further a scenario into a leading candidate, is largely a question of being able to play the game in which it is crucial to give language and shape to possible futures that resonate with both managers and colleagues, hence being futures already (partly) sensed and anticipated in the organization. The arational techniques of enchantment, in other words, function as strategies that shape ideas and visions of others by way of appeal and persuasion.

Conclusion: the ontological implications of prefiguration

In this article, we have proposed the concept of technologies of prefiguration to capture how organizational actors seek to bring the future into the present in order to act upon it. Based on an ethnographic analysis of the strategy of scenarios in the medical company MedTech, we argue that establishing the 'presence' and 'rightness' of what may, or may not, happen is fundamentally a performative operation which, although being outward oriented, is embedded in a political economy of futures, amounting to an organizational game with various techniques of enchantment playing crucial roles. While technologies of prefiguration thus render strategic actions into indeterminate futures possible, they also have internal organizational effects in terms of cultivating particular organizational practices, subject positionings, and work cultures.

Here, we place emphasis on how the performative enactment of technologies of prefiguration have significant ontological implications. The technologies appear to be fuelled by a distinct conundrum of late modernity: on the one hand, the future figures as an open horizon for human intervention while, on the other, it is essentially unknowable and will, as Strathern (1992) remarks, determine its own genealogy (172). For actors in organizations, the future thus represents a problematic temporal category that cannot be mastered through (rational) planning practices alone (Wenzel et al. 2020). Rather, as our ethnography shows, the future becomes a game, or even a battlefield, for multiple and competing alternatives. In a logic of defuturing, an exploration of these alternatives may inform decisions and actions, but the proliferation of possible futures does not eliminate uncertainty 'about the appropriateness of organizational activities to be performed in the present' (Wenzel et al. 2020, 1450). Instead, various practices, alliances, and techniques may gain significance in the shadows of the formal processes, rendering it pertinent for organizational researchers and professionals to uncover the situated and heterogeneous – rational as well as arational – ways in which futures are prefigured and fashioned.

This is important. For technologies of prefiguration are not mere innocent and playful endeavours of creative imagination. To prefigure is to establish the *presence* of an alternative future, meaning that it is, in fact, already in the making. As a technology, in a Foucauldian sense, this presence impacts and shapes the perceptions and actions of organizational actors, not least in terms of their perspective on the outside world, its opportunities and threats, and how these may be acted upon. In brief, technologies of prefiguration infiltrate organizational agendas and are, thus, constitutive of important parts of organizational life. This is why they are inherently ontological, bringing about particular ways of knowing, thinking, and acting that foreground and authorize certain future worlds over others. While highly contested, futures are always already here.

Notes

1. The names of MedTech, departments, managers, and employees in this article are all pseudonyms.
2. See Skoglund and Böhm (2020) for a more elaborate review of prefigurative politics in organizational research (1261-1263).
3. As an analytical concept, technologies of prefiguration bears similarities to the concept of 'technologies of the imagination', which Sneath, Holbraad, and Pedersen (2009) put forward as a way to capture 'the social and material means by which particular imaginings are generated' (6). However, while imagination is part of future-making in organizations (Beckert 2021), technologies of prefiguration designate more organized and purposeful knowledge processes which are intended to foretell and prefigure the future (or futures) by giving it (or them) a specific form and content that allow for exploration, scrutinization, discussion, and decision-making.
4. Such studies include a focus on pandemic preparedness measures (Lakoff 2008; Samimian-Darash 2009), insurance and resilience rebuilding policies against natural hazards (Collier, Cox, and Grove 2016), civil defence planning against nuclear attacks (Collier 2008), and national governments' enactment of scenario-based exercises to prepare for events 'whose probability cannot be calculated, but whose consequences are potentially catastrophic' (Lakoff 2008, 403; Samimian-Darash 2016).
5. 'In our endeavor to sustain ourselves in the short term', Fry (2009) writes, 'we collectively act in destructive ways towards the very things we and all other beings fundamentally depend upon' (22), thus making such destructive acts essentially 'defuturing'. Hunt (2011) elaborates that the concept also concerns how all design 'is an act both of prefiguring a future social milieu but also *the erasure of multiple possible alternatives* […] It is an ontology of prefigurement that destroys as it creates' (35, emphasis added).
6. In organization studies, discussions of enchantment have generally aimed to move beyond Weber's dichotomy between scientific rationality and spiritual enchantment (Hancock 2005). The discussions have, however, been divided by two distinct analytical perspectives, construing enchantment as either a product of human intervention or an expression of an 'authentic' search for meaning. See Mauksch (2017, 137-139) for a more detailed discussion of these positions.

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